12.8 Solids of Revolution

Geometry 12.8 · Student sheet

Name _____ Period ____ Date _____

I can…

- name the solid formed by rotating a figure around an axis.
- tell which measures become the radius and the height.
- find the volume of a solid of revolution.

Vocabulary · fill in the missing word

Solid of revolution — a _____ figure formed by rotating a two-dimensional shape around an axis

Axis of revolution — the line around which the shape is _____

Part 1 · Discovery: spin a flag

Tape each shape to a pencil along one side and spin the pencil. Name the solid each one makes.

Shape	Taped along	Solid it makes	Radius comes from	Height comes from
Rectangle	one side			
Right triangle	one leg			
Semicircle	its diameter			—

The side along the pencil becomes the _____.

Part 2 · Notes and examples (fill in each note, then work the example)

Basic solids of revolution: a rectangle rotated around one side makes a _____; a right triangle rotated around a leg makes a _____; a semicircle rotated around its diameter makes a _____.

Example 1

Name the solid formed by rotating each figure around the given axis.

a. a rectangle around one of its sides

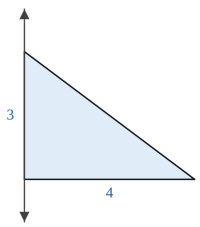
b. a right triangle around one of its legs

c. a semicircle around its diameter

Radius and height: the side on the axis becomes the _____; the side perpendicular to the axis becomes the _____.

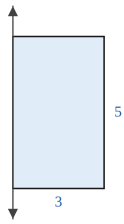
Example 2

The right triangle is rotated around the leg on the axis. Describe the solid: name it and give its radius and height.



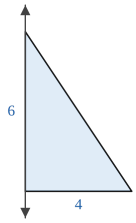
Example 3

The rectangle is rotated around the side on the axis. Find the volume of the solid. Round to the nearest hundredth.



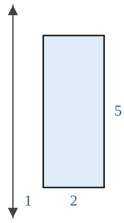
Example 4

A right triangle with legs of 6 (on the axis) and 4 is rotated around the axis. Find the volume of the solid. Round to the nearest hundredth.



Example 5

A 2-by-5 rectangle sits 1 unit away from the axis, as shown, and is rotated around it. Find the volume of the solid. Round to the nearest hundredth.



Part 3 · Partner practice

Partner 1

A 4-by-7 rectangle is rotated around its side of length 7. Find the volume of the solid. Round to the nearest hundredth.

Partner 2

A semicircle with a radius of 3 is rotated around its diameter. Find the volume of the solid. Round to the nearest hundredth.